MIL-D-23615B(AS)
5 March 1974
Superseding
MIL-D-23615A(AS)
15 June 1966

MILITARY SPECIFICATION

DESIGN AND EVALUATION OF CARTRIDGE ACTUATED DEVICES

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the general requirements for design and establishes uniform methods for testing cartridge actuated devices (CADs). Guns, rockets, rocket catapults, ballistic hose/tubing, and stores separation equipment are not covered by this specification. For purposes of this specification, the term cartridge actuated device shall include any device which provides the means of releasing potential cartridge energy or harnessing this energy to accomplish work. Linear shaped charge, explosive energy transfer lines, and "sealed-in" cartridge actuated devices (see 6.3.1) shall fall within the scope of MIL-D-21625. However, devices of the "sealed-in" type shall also be required to meet the applicable safety and structural design requirements of this specification as specified in 3.6.3. The purpose of the testing program is to determine performance, safety, soundness of mechanical design, and resistance to environments encountered during storage, handling, and service use. Cartridge actuated devices must be safe for handling and must not deteriorate to a degree which would render their performance or safety doubtful after being subjected to the applicable testing programs of this specification.

1.2 Classification of the types of cartridge actuated device release.

Type I

Safety of handling and installation. This type of release is required prior to handling and installation of CADs in naval equipment. For requirements, see 3.16.1.

Type II

Interim service release. This release is required prior to any flights or use of CADs involving naval personnel and will be granted in accordance with the provisions of 3.16.2. The Naval Air Systems Command has the prerogative to authorize procurement of CADs for fleet use, subsequent to a Type II release.

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Type III A

Final release to service (fully documented). This type of release is required for admission of CADs into the naval supply system as a fully documented device (see 3.9). For requirements, see 3.16.2.

Type III B

Special release to service (documented by a Source Control Drawing). If the data required for a Type III A release are not available (i.e., proprietary rights, patents, etc.) then a Type III B release to service is required for admission of CADs into the naval supply system. For requirements for a Type III B release see 3.16.2. For documentation and data requirements for a Type III B release, see 3.9.

Type IV

Use of approved cartridge actuated device (using same ballistic energy source and associated equipment) in a new application. If a CAD has previously been granted a Type III A or Type III B release and is to be used in a new application, a Type IV release to service for the new application will be granted in accordance with the requirements of 3.16.3.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-P-514	Plate, Identification, Instruction and Marking, Blank
MIL-D-1000	Drawings, Engineering and Associated List
MIL-S-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-C-5501	Caps and Plugs, Protective, Dust and Moisture Seal
MIL-C-5541	Chemical Conversion Coatings of Aluminum and Aluminum Alloys

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MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-P-19834	Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-23460	Pin, Quick-Release, Self-Retaining, Positive Locking
MIL-C-83124	Cartridge Actuated/Propellant Actuated Devices, General Design Specification For

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-875	Type Designation System for Aeronautical and Aeronautical Support Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1166	Radiographic Testing Requirements for Solid Propellants

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Navy Department

WR-43A	Preparation of Quality Assurance Provisions

National Aerospace

NAS 618	Fastener-Recommended Shank, Hole and Head-to-Shank Fillet Radius Limits for
NAS 1091	Streamer Assembly, Warning

PUBLICATIONS

DSM 4120.3-M	Standardization Policies, Procedures and Instructions
49 CFR 171-178	Transportation, Explosives and Other Dangerous Articles

National Bureau of Standards

Handbook H-28	Screw-Thread Standards for Federal Service

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS.

3.1 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2 Special requirements. Special requirements for specific applications shall take precedence over those listed herein, provided such special requirements are more stringent. Other conflicting requirements are subject to the approval of the cognizant design agency for the specific application involved.

3.3 Materials. Materials for the CAD shall be compatible with the explosive, propellant, delay composition, pyrotechnic, combustion products, and buffering fluid (if used); and shall withstand environmental, functional, service, and storage conditions to which the CAD will be exposed. Acceptance or approval of materials for design or use during the course of manufacture shall in no case be construed as a guarantee of acceptance of the finished device. Pyrocellulose materials shall not be included as wads, spacers, or closure discs. Materials which are nutrients for fungi shall not be used.

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3.3.1 Metals. Exposed metals shall be of the corrosion-resisting type or suitably treated to resist the corrosive effects of fuels, salt spray or atmospheric conditions to which the CAD may be subjected in storage or normal service use.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion. Dissimilar metals are defined in MIL-STD-889.

3.3.2 Plastic parts. The use of plastic parts shall be subject to the approval of the cognizant design agency for the specific application involved.

3.4 Finishes. Protective coatings and finishes shall be used which will not crack, chip, or scale during normal service life or when subjected to environmental conditions specified herein. Surface treatments, coatings, and finishes shall conform to MIL-S-5002 except that aluminum and aluminum alloy parts shall be anodized in accordance with 3.4.1.

3.4.1 Anodizing. Aluminum and aluminum alloy parts subject to wear, abrasion, and erosion or exposed to corrosive environmental elements shall be anodized in accordance with MIL-A-8625, Types II or III. Chemical conversion coatings conforming to MIL-C-5541 may be used to repair mechanically damaged areas from which the anodic coating has been removed.

3.5 Color coding. Color coding of CADs for identification purposes is strictly forbidden.

3.6 Design and construction. The CAD shall be of the simplest and most efficient design consistent with the proposed use. Specific consideration shall be given to the factors of minimum size and weight, resistance to deleterious environments, reliability of performance, maintainability (see 3.6.2), and safety of operation. All CADs including the ballistic energy source, i.e., cartridges, explosive energy transfer lines, etc., submitted for or subjected to the requirements of 3.16 shall be of final design configuration, identical in design, manufactured by the same process and shall be from the same lot.

3.6.1 Ballistic energy. Prior to service release, the cartridge(s) for the CAD shall have met the applicable release requirements of MIL-D-21625.

3.6.2 Disassembly and reassembly. Unless otherwise approved by the cognizant design agency, the CAD shall be designed in such a manner that it can be disassembled for inspection or installation of the cartridge(s),

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"O" rings, and associated subassemblies as applicable and reassembled. Disassembly and reassembly of the device shall be accomplished by the use of standard tools.

3.6.3 "Sealed-in" type of cartridge actuated device. "Sealed-in" type of cartridge actuated devices (see 6.3.1) shall meet the design, environmental and functional requirements of MIL-D-21625 and the firing mechanism, shear pin, dust, submersion, iced condition and structural integrity requirements of this specification.

3.6.4 Service life. CADs when packaged in containers shall have a service life of not less than five years from the date of manufacture, and shall be capable of being rendered serviceable after this period by replacement of cartridges, "O" rings, shear pins, and similar items. Service life assignments must be approved by the Naval Air Systems Command prior to application in service.

3.6.5 Structural integrity. CADs shall be designed with a minimum safety factor of 1.5 at the most critical temperature extreme based on the maximum calculated operating pressure limit in the most severe operating condition.

3.6.5.1 Locked shut firing. Each CAD shall be designed to withstand the internal ballistic gas pressure developed during a locked shut condition (see 6.3.4) over the temperature range -65°F to +200°F. There shall be no mechanical failure or rupture.

3.6.5.2 No load firing. For a stroking CAD which is designed to remain intact at the end of a power stroke, parts shall not separate when it is fired at any temperature between -65°F and +200°F with no restraining load applied to the CAD, e.g., the piston shall not separate from the CAD at the end of stroke nor shall fragmentation of components occur.

3.6.6 Locking of parts. All parts not required to have relative motion prior to or during cartridge/ballistic energy source initiation shall be securely locked by a method that will not damage components or interfere with ease of disassembly. Safety wiring is the preferred method. Lock nuts shall not be used except where it is impossible to use the preferred method. Floating or press fit washers shall not be used in the CAD.

3.6.7 Mechanical firing mechanism. The force required to actuate mechanical firing mechanisms shall be 15 pounds minimum and 35 pounds maximum. Precocked firing mechanisms shall not be used (see 6.3.2).

3.6.7.1 Safety pins. Mechanical firing mechanisms shall be restrained by single acting safety pins which meet the requirements of MIL-P-23460

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(Type I, 1/4 inch minimum diameter) unless otherwise specifically approved by the cognizant design agency. The longitudinal axis of the installed safety pin shall be 90 degrees to the direction of travel of the restrained member. The safety pin shall act in double shear. Interference between the safety pin and the restrained member shall be 0.062 inch minimum. Removal and reinsertion of the safety pin shall be accomplished after the mechanical firing mechanism is securely locked in place (see 3.6.6). The pin length selected will be dependent upon the size and configuration of the device and shall provide for positive locking of the pin. Safety pin hole diameters (in the device) shall be within the hole limits for clearance fits specified by column C of NAS 618. Safety pins shall be accessible and visible when the device is installed and shall not interfere with installation or removal of the device or any other maintenance operation.

3.6.7.2 Warning streamers. Warning streamers designed in accordance with NAS 1091 shall be securely attached to all safety pins. Streamer length shall be as specified by the cognizant design agency.

3.6.8 Gas actuated firing mechanisms. The firing mechanism shall not actuate upon application of 400 psig gas pressure on the firing pin. The firing pin retention mechanism, e.g., shear pin(s), etc., shall release within 0.030 second upon application of 600 psig applied at a rate between 10,000 psig/sec and 50,000 psig/sec inclusive acting on the firing pin assembly. Time to actuate is only a design guide and shall be reviewed by the cognizant design agency for each specific application.

3.6.9 Shear pins. Shear pins employed within CADs shall be of a material which demonstrates shear strength characteristics within the specified tolerance over the temperature range -65°F to +200°F.

3.6.10 Installation. Installation provisions, e.g., mounting holes, flanges, brackets, etc., for CADs shall be so arranged that the possibility of incorrect installation or connection of fittings is minimized. Utilization of male inlet/outlet ports is preferred. Use of inlet/outlet ports of different thread diameters in mandatory.

3.6.11 Screw threads. All screw threads shall be specified in accordance with National Bureau of Standards Handbook H-28.

3.6.12 Cycle life. CAD component(s) which experience relative motion prior to cartridge/ballistic energy source initiation shall have a service cycle life. As a minimum, the number of cycles a CAD component(s) shall withstand would be comparable to its normal service cycles plus a 50 percent cycle safety factor. Service cycle life shall be subject to the approval of the cognizant design agency. A cycle shall consist of movement from the initial position through all positions and back to the initial position.

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3.6.13 Actuation. For CADs which utilize ballistic energy from an upstream source to accomplish work, the ballistic input at the point of actuation shall provide a minimum safety factor of 1.5 based on the required CAD actuation pressure under worst case condition. Operational time shall be as specified in 3.6.8.

3.7 Nomenclature. The contractor shall follow the procedure outlined in MIL-N-18307 and MIL-STD-875 when submitting the DD Form 61 for nomenclature. The nomenclature requested by the contractor on DD Form 61 shall be concurred in by the Naval Air Systems Command prior to use.

3.8 Marking. Identification marking for each CAD shall be clearly and permanently (nondefaceable through normal storage and service handling) marked in accordance with MIL-STD-130 with the CAD nomenclature (see 3.7), manufacturer/serial number, contract number, lot number, manufacturer's symbol or federal stock number/federal supply code number and date of manufacture. Name plates shall conform to MIL-P-514 requirements for composition A, Class 2 or composition C identification plates, or MIL-P-19834. No other markings shall be placed on the CAD except those required by 3.8.1 and 3.8.2.

3.8.1 Inlet/outlet ports. CAD ports shall be labeled "inlet" or "outlet", as applicable, and the direction of flow indicated with permanent red arrows. In addition, ports shall be covered with suitable and effective shipping caps or plugs in accordance with MIL-C-5501, type and/or class as applicable, when the CAD is not installed.

3.8.2 Warnings. Red tags or labels in accordance with MIL-P-19834 with "WARNING - CARTRIDGE ACTUATED DEVICE (INCLUDING SAFETY INSTRUCTIONS)" lettered in white shall be provided for and affixed to all CADs.

3.9 Contractual documentation and data requirements. The cognizant government contracting agency when preparing a contract which includes CAD development shall include, as a minimum, on the DD Form 1423 the following data requirements:

3.9.1 Reports in accordance with MIL-STD-831 of all tests performed by the prime or subcontractor during development (see 3.15) and qualification of the CAD and any statistical analysis of these results.

3.9.2 Drawings including special tooling required in accordance with MIL-D-1000 Category E, Form 1. Also packaging and packing drawings to supplement 3.9.5.1 of this specification.

3.9.3 If full disclosure cannot be obtained for Category E, Form 1 drawings in accordance with MIL-D-1000 then a source control drawing, as defined by MIL-D-1000 Category F, Form 3 shall be submitted. Also the source control drawing shall include packaging and packing information in sufficient detail to meet the requirements of Section 5 of this specification. If Category F, Form 3 drawings are submitted, complete drawings containing restrictive use notation shall be supplied to facilitate in-service support/malfunction investigation.

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3.9.4 Detail and assembly drawings of test sets, including special tooling required, in accordance with MIL-D-1000 Category E, Form 3 and procedures used during testing, to permit manufacture and use of identical test sets for follow on production testing.

3.9.5 Design, production and acceptance specifications in accordance with Defense Standardization Manual DSM 4120.3-M.

3.9.5.1 Packaging and preservation procedures. See 6.4.

3.9.6 Classification of characteristics in accordance with WR-43A.

3.9.7 Complete manufacturing drawings of the CAD and of test sets used in the development and evaluation/qualification of the CAD should be submitted to the cognizant design agency concurrently with or prior to the submittal of the test report.

3.9.8 The contractor shall prepare the above data, drawings, specifications, etc., for cognizant design agency approval.

3.10 Performance. The CADs shall satisfy all design, environmental, and functional requirements specified herein and reliability and performance requirements outlined in the detailed CAD design specification.

3.10.1 Radiographic examination. Each CAD shall be subjected to radiographic examination such as X-ray, neutron bombardment, gamma ray, etc., as specified in 4.5.2. Use of radiographic techniques not covered by military specification or previous approval by the cognizant design agency shall be submitted for approval. Radiographic plates shall be identified by date, CAD lot number, and serial number. Radiographic safety procedures shall be in accordance with MIL-STD-1166. Radiographic plates shall be retained until acceptance of the qualification test report by the cognizant design agency.

3.10.2 Dimensions. Unless otherwise specified, dimensions shall apply after all machining, process treatments (plating, anodizing, heat treating), and nondestructive testing have been completed. No part of the assembly shall deviate from the drawing configuration, dimensions, and tolerances.

3.11 Environmental conditions.

3.11.1 Forty-foot drop. To ensure safety of handling, the CAD shall not fire when dropped from a height of 40 feet as specified in 4.6.1 and shall be safe for handling and disposal.

3.11.2 Six-foot drop. The CAD shall not fire when dropped from a height of six feet as specified in 4.6.2 to ensure safety of handling and installation. After being subjected to the six-foot drop test, the CADs shall meet the design performance requirements when test fired.

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3.11.3 Shock. The CADs shall withstand shock conditions as specified in 4.6.3. After subjection to the shock test, the CADs shall meet the design performance requirements when test fired.

3.11.4 Temperature and humidity cycling and storage. The CADs shall withstand the temperature and humidity cycling and storage conditions as specified in 4.6.4 and shall meet the design performance requirements when test fired.

3.11.5 Vibration. The CADs shall withstand vibration conditions as specified in 4.6.5. After subjection to the vibration test, the CADs shall meet the design performance requirements when test fired.

3.11.6 Salt fog. The CADs shall withstand exposure to salt fog as specified in 4.6.6 and shall meet the design performance requirements when test fired.

3.11.7 Dust. The CADs shall withstand exposure to dust as specified in 4.6.7. After subjection to the dust test, the CADs shall meet the design performance requirements when test fired.

3.11.8 Cycling. The CADs shall withstand cycling conditions as specified in 4.6.8 and shall meet the design performance requirements when test fired.

3.12 Functional tests.

3.12.1 Operation at -65°F and sea level pressure. See 4.7.2, 4.7.5, and 4.7.6.

3.12.2 Operation at +200°F and sea level pressure. See 4.7.3, 4.7.5, and 4.7.6.

3.12.3 Operation at +70°F and sea level pressure. See 4.7.4, 4.7.5, and 4.7.6.

3.12.4 Operation at 80,000 ft altitude and -65°F. See 4.7.7.

3.12.5 Submerged operation. See 4.7.8.

3.12.6 Operation under iced condition. See 4.7.9.

3.13 Structural integrity tests.

3.13.1 Locked shut operation at -65°F and +200°F, at sea level pressure. See 4.7.10.1.

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3.13.2 No load operation at -65°F and +200°F, at sea level pressure. See 4.7.10.2.

3.14 Damage and deterioration. Damage to or deterioration of any internal or external part of the CAD which could in any manner prevent it from meeting operational requirements shall be reason to consider the CAD as having failed to meet the test to which it was subjected.

3.15 Development testing. Development tests shall be conducted to provide data to be used in the design freeze of the CAD. Development tests may be used to determine operating characteristics. Stress to failure tests shall be conducted during development to provide failure mode characteristics for verification of analysis and determination of strength and operating margins of safety. It is required that development tests be run on the CAD under simulated thermal environments to assure design compatibility with the requirements of this specification. Vibration amplification factors shall be substantiated during development testing. Development tests shall be categorized as follows:

(a) Design feasibility tests (3.15.1).

(b) Design verification tests (3.15.2).

3.15.1 Design feasibility tests. Prior to DFT (design feasibility tests), conceptual drawings and critical supporting calculations shall be reviewed by the cognizant design agency to determine compliance with the requirements specified herein. Design feasibility tests shall include all tests conducted for the following purposes:

(a) Component and part selection.

(b) Investigation of development model performance shall include as a minimum, functional test performance to indicate rupture, operating margins, and structural integrity.

(c) Demonstration of safety and operating margins or other analytical assumptions (see 3.6.5).

3.15.2 Design verification tests. DVT (design verification tests) are advanced stage development tests which are performed on the CADs for the purpose of substantiating the design for its intended mission. These tests shall include as a minimum, radiographic inspection, the tests listed in Table I, humidity tests, and certification of 3.15.1. Successful completion of DVT and concurrance and approval by the cognizant design agency provides the assurance to permit a design freeze decision to be made.

3.16 Criteria of acceptance.

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3.16.1 Minimum test requirements. After successful completion of DVT (see 3.15.2), minimum requirements for a Type I release are satisfactory performance in the tests listed in Table I as performed by the government, contractor, or an independent testing laboratory. If the requirements of 3.15.2 are met without design change, the results may be accepted in fulfilling the requirements for a Type I release. If the tests are performed by a contractor or by an independent testing laboratory, the test facility must be approved by the cognizant design agency and the tests witnessed by government personnel. A certified copy of the test data shall be submitted to the cognizant design agency for approval prior to granting of a Type I release.

TABLE I

Test	Number of CADs Required	Applicable Section for Test Procedures	Discard	Functional Test at Indicated Temperature
Visual	36	4.5.1		
Radiographic Inspection	36	4.5.2		
Expend to test listed below				
40-Foot Drop(1)	6	4.6.1	6	
6-Foot Drop	3	4.6.2		3 @ +70°F
Shock	4	4.6.3		4 @ +70°F
Vibration	4	4.6.5		4 @ +70°F
Low Temperature (-65°F)	3	4.7.2		3 @ -65°F
High Temperature (+200°F)	3	4.7.3		3 @ +200°F
Normal Temperature (+70°F)	3	4.7.4		3 @ +70°F
Firing Mechanism Mechanical/Gas Actuated	3	4.7.5.1/ 4.7.5.2		1 @ -65°F 1 @ +70°F 1 @ +200°F
Actuation	3	4.7.6		1 @ -65°F 1 @ +70°F 1 @ +200°F
Locked Shut	2	4.7.10.1		1 @ -65°F 1 @ +200°F
No-Load	2	4.7.10.2		1 @ -65°F 1 @ +200°F

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(1) Inert CADs without explosive or propellants will not be subjected to this test.

3.16.2 A Type II, Type III A or Type III B release to service can be given only after the CAD satisfies the environmental, functional, material, design and construction, and documentation and data requirements, and after successful performance in the complete testing program as set forth in Table II. If the tests are to be performed by a contractor or by an independent testing laboratory, the test facility must be approved by the cognizant design agency and the tests witnessed by government personnel. A certified copy of the test data shall be submitted to the cognizant design agency for approval prior to granting of a Type II, Type III A or Type III B release to service. Applicable documentation for either a Type III A or Type III B release in accordance with the requirements of 3.9 shall be prepared as determined by contractual agreement and submitted along with the test data. Distribution of CADs for the various environmental and functional tests shall be in accordance with Table II. CADs which have met the requirements of MIL-C-83124 may be considered acceptable for a Type II, Type III A, or Type III B release provided all applicable requirements of this specification have been met.

TABLE II

Test(1)	Number of CADs Required	Applicable Section for Test Procedures	Discard	Functional Test at Indicated Temperature
Visual	100	4.5.1		
Radiographic Inspection	100	4.5.2		
Expend to test listed below				
40-Foot Drop(2)	6	4.6.1	6	
6-Foot Drop	3	4.6.2		3 @ +70°F
Shock	4	4.6.3		4 @ +70°F
Temperature and Humidity Cycling and Storage	12	4.6.4		12 @ +70°F
Vibration	4	4.6.5		4 @ +70°F
Salt Fog	4	4.6.6		4 @ +70°F
Dust	4	4.6.7		4 @ +70°F
Cycling	2	4.6.8		2 @ +70°F

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TABLE II (Cont'd)

Test(1)	Number of CADs Required	Applicable Section for Test Procedures	Discard	Functional Test at Indicated Temperature
Low Temperature (-65°F)	2	4.7.2		2 @ -65°F
High Temperature (+200°F)	2	4.7.3		2 @ +200°F
Normal Temperature (+70°F)	2	4.7.4		2 @ +70°F
Firing Mechanism Mechanical/Gas Actuated	6	4.7.5.1/ 4.7.5.2		2 @ -65°F 2 @ +70°F 2 @ +200°F
Actuation	6	4.7.6		2 @ -65°F 2 @ +70°F 2 @ +200°F
High Altitude	4	4.7.7		4 @ -65°F
Submersion	4	4.7.8		4 @ Ambient
Iced Condition	4	4.7.9		4 @ -65°F
Locked Shut	2	4.7.10.1		1 @ -65°F 1 @ +200°F
No-Load	2	4.7.10.2		1 @ -65°F 1 @ +200°F

(1) If the tests of Table II are performed by a contractor as an independent testing laboratory the 27 devices that were not allocated to the destructive test of Table II are to be furnished to a Navy activity designated by the procuring agency. These devices will be used to verify production lot acceptance procedures and test set functioning as being appropriate for acceptance and follow-on support for production lot procurements. If the test of Table II are performed by a government laboratory the devices that were not allocated to the destructive tests of Table II shall be retained for investigative purposes.

(2) Inert CADs without explosive or propellants will not be subjected to this test.

3.16.3 Acceptance of an approved CAD (using same ballistic energy source and associated equipment) in a new application. When it is desired to use a previously approved CAD (which has a Type III A or Type III B release) in a new application, Type I service release is automatically granted. A

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Type IV release for service use will be granted upon satisfactory performance in any special tests required by the new application (see 3.2) and the tests of Table II deemed appropriate for the new application by the cognizant design agency.

3.17 Packaging, packing and marking. The packaging and packing of CADs and the container markings shall be in accordance with Section 5. See 6.4.

3.18 Workmanship. Workmanship shall be subjected to the inspection of all requirements as specified herein and as specified in 3.9 as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Materials. Inspection and testing of component parts and assemblies shall be made to determine compliance with 3.3. Where defects or inferior quality are evident, and the Government deems material analysis necessary, the contractor will be requested to submit samples or specimens to the cognizant design agency for analysis and approval.

4.3 Measurements and records. All functional tests of CADs shall be performed with the equipment mounted and loaded in a manner to simulate service conditions as closely as possible. Applicable performance parameters such as the following shall be recorded during functional tests:

a. Pressure versus time for ballistic cycle time.

b. Velocity (at end of power stroke).

c. Displacement of load (or simulated) versus time.

d. Thrust versus time for ballistic cycle time.

e. Acceleration versus time.

f. Resistive load versus time.

g. Time interval between actuating firing mechanism and beginning of movement of load or start of pressure rise.

h. Time interval between actuating firing mechanism and completion of operating cycle.

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4.3.1 Instrumentation. Any state-of-the-art type of instrumentation and recording equipment may be used, e.g., oscillograph, oscilloscope, or magnetic tape. Transducers may be any state-of-the-art type, e.g., piezoelectric, strain gage, variable reluctance, capacitive, or potentiometer. However, accuracy of all test apparatus shall conform to requirements of MIL-STD-810. Documented calibration records shall be maintained and be available for inspection by the cognizant design agency.

4.4 Rejection and retest. Failure of any CAD subjected to the environmental and functional test to conform to the applicable requirements of this specification or the predetermined critical design requirements may be cause for rejection of the entire lot (see 3.6). Full particulars concerning the failure, recommended action to correct the defect(s), and plan for retest shall be submitted to the cognizant design agency for approval. The retest must be approved by the cognizant design agency.

4.5 Non-destructive tests. Individual tests shall be conducted on each CAD submitted for environmental and functional test. If any device fails a part of the non-destructive test, the cognizant design agency may reject the entire lot (see 3.6). Following concurrance and approval by the cognizant design agency, the contractor shall correct all deficiencies prior to resubmitting the rejected articles for non-destructive retest. Non-destructive tests shall consist of those specified in 4.5.1 and 4.5.2.

4.5.1 Examination of product. Each CAD shall be completely inspected for compliance with the requirements specified herein, workmanship (see 3.18), and its applicable drawings and specification requirements (see 3.9) prior to environmental and functional testing.

4.5.2 Radiographic inspection. All CADs shall be inspected by radiographic means as specified in 3.10.1 and plates examined for defects and misassembly.

4.6 Environmental tests. All temperatures specified in 4.6.2 through 4.6.8 shall have a tolerance of not more than ± 5°F. If the CADs cannot be test fired within the temperature chamber, they shall be fired within five minutes after removal from the chamber. Whenever it is necessary to repeat low (-65°F) temperature conditioning of a cold device, all condensation shall be removed from the CAD before it is returned to the temperature conditioning chamber.

4.6.1 Forty-foot drop test. This test shall be conducted in accordance with test 103 of MIL-STD-331. Six CADs shall be dropped to impact in the following positions: (1) two nose up, (2) two nose down, and (3) two horizontal. A new CAD shall be used for each drop. The CADs shall meet the requirement of 3.11.1.

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4.6.2 Six-foot drop test. Three CADs shall be dropped onto a 2-inch thick steel plate (minimum) embedded in concrete to impact as follows: (1) one nose up, (2) one nose down, and (3) one horizontal. A new CAD shall be used for each drop. No device shall fire during this test. Subsequent to the drop test, the CADs shall be test fired at +70°F.

4.6.3 Shock. This test shall be conducted in accordance with method 516, procedure 1 of MIL-STD-810. The shock pulse wave form shall be terminal peak sawtooth. The peak amplitude shall be 20g and the duration shall be 11 msec. After being subjected to the shock test, the CADs shall be test fired at +70°F.

4.6.4 Temperature and humidity cycling and storage. This test provides for cycling between temperatures of -65°F, +70°F and +160°F (90 percent relative humidity) with additional storage at -80°F and +160°F for two periods of two days each. Provision has been made for withdrawal of CADs at two times during the cycling (see schedule below). This is to provide an opportunity to observe the extent of progressive deterioration, if such exists. The schedule has been arranged in such a manner that operations are not required outside regular working hours except for such supervision as may be necessary to insure proper operation of the controlled temperature cabinets. CADs shall be supported on screen trays so that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test. CADs shall be free of visible damage and shall produce satisfactory ballistic performance when test fired. CADs will be conditioned at +70°F for functional tests. The schedule to be followed is:

Monday	1300	Place in cabinet or room maintained at -65°F;
	1600	Remove from -65°F room and place in room maintained at +160°F, (90% relative humidity) and allow to remain overnight;
Tuesday	0800	Remove from +160°F and place in +70°F;
	1300	Remove from +70°F and place in -65°F;
	1600	Remove from -65°F and place in +160°F (90% relative humidity);
Wednesday	0800	Remove from +160°F and place in +70°F;
	1300	Remove from +70°F and place in -65°F;
	1600	Remove from -65°F and place in +160°F (90% relative humidity);

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Thrusday	0800	Remove from +160°F and place in +70°F;
	1300	Remove from +70°F and place in -65°F;
	1600	Remove from -65°F and place in +160°F (90% relative humidity);
Friday	0800	Remove from +160°F and place in +70°F;
	1300	Remove from +70°F and place in -65°F;
	1600	Remove from -65°F and place in +160°F (90% relative humidity);
Saturday and Sunday		Maintain in +160°F (90% relative humidity);
Monday	0800	Remove from +160°F and place in +70°F;
	1300	Remove from +70°F and place in -65°F;
	1600	Remove from -65°F and place in +160°F (90% relative humidity);
Tuesday	0800	Remove from +160°F and place in +70°F;
	1300	Remove from +70°F and place in -65°F;
	1600	Remove from -65°F and place in +160°F (90% relative humidity);
Wednesday	0800	Remove from +160°F and place in +70°F;
	1300	Remove from +70°F and place in -65°F;
	1600	Remove from -65°F and place in +160°F (90% relative humidity);
Thursday	0800	Remove from +160°F and place in +70°F;
	1300	Remove from +70°F and place in -65°F;
	1600	Remove from -65°F and place in +160°F (90% relative humidity);
Friday	0800	Remove from +160°F and place in +70°F;
	1300	Remove from +70°F and place in -80°F;

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Saturday and Sunday		Maintain at -80°F;
Monday	0800	Remove from -80°F and place in +70°F
		Remove four CADs. Condition two at +70°F and fire.
		Store remaining two CADs (at +70°F) which are to be fired at end of cycling program.

The second two-week period follows the same schedule of temperature and humidity cycling and storage. Withdrawals of CADs during the second two-week period are to be as follows:

Fourth Monday withdraw four CADs, condition two at +70°F and fire.

Store remaining two CADs (at +70°F) which are to be fired at end of cycling program.

The completion of the two periods occurs at 0800 on the 5th Monday after starting. At this time, the remaining four CADs are to be removed, conditioned at +70°F, together with the remainder of CADs from two and three weeks cycling, and fired.

4.6.5 Vibration. This test shall be conducted in accordance with the vibration test selection chart in Table 514.1-I using curve Z of MIL-STD-810, except that for each resonant and cycling period the test specimens shall be vibrated as follows: (1) one at -65°F, (2) two at +70°F, and (3) one at +200°F. After vibration testing has been completed, the CADs shall be test fired at +70°F.

4.6.6 Salt fog. This test shall be conducted in accordance with method 509 of MIL-STD-810. After the salt fog test has been completed and without disassembling the devices, the external surfaces may be rinsed with tap water prior to test firing the specimens at +70°F.

4.6.7 Dust. This test shall be conducted in accordance with method 510 of MIL-STD-810 except that the high temperature shall be +200°F. After the dust test has been completed, the CADs shall be test fired at +70°F.

4.6.8 Cycling. Two CADs shall be subjected to the cycling test by moving the CAD component(s) from the initial position, through all positions, and back to the initial position. The cycling test shall be conducted at +70°F. The number of cycles (see 3.6.12) and cyclic rate shall be specified in the detailed CAD design specification. Subsequent to the cycling test, the specimens shall be test fired at +70°F.

4.7 Functional tests.

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4.7.1 General. Operations at extreme temperatures and under high altitude conditions will be performed where possible in a controlled condition room or chamber utilizing the energy source(s) and associated equipment that will be used in service. CAD physical size and amount of travel will generally be the limiting factors. All temperature specified in 4.7.2 through 4.7.10 shall have a tolerance of not more than ± 5°F. Test data shall be recorded as applicable. See 4.3.

4.7.2 Low temperature tests (-65°F). The CADs and equipment are to be conditioned at a temperature of -65°F for a period of one hour longer than that required to reach thermal equilibrium and then fired. When necessary, the test specimens may be removed from the conditioning chamber, set up, and fired within a period of five minutes.

4.7.3 High temperature tests (+200°F). The CADs are to be fired at a temperature of +200°F. Conditioning time and method of firing shall be as specified in 4.7.2.

4.7.4 Normal temperature tests (+70°F). The CADs are to be fired at a temperature of +70°F. Conditioning time and method of firing shall be as specified in 4.7.2.

4.7.5 Firing mechanism tests.

4.7.5.1 Mechanical firing mechanism. For CADs employing a mechanical firing mechanism, the rounds shall be fired as follows: (1) one-third at -65°F, (2) one-third at +70°F, and (3) one-third at +200°F. Conditioning time and method of firing shall be as specified in 4.7.2. In addition to the requirements of 4.3, the firing mechanism pull force shall be recorded. Force values shall be as specified in 3.6.7. Direction of pull shall be "in line" with the axis of the firing mechanism.

4.7.5.2 Gas actuated firing mechanism. For CADs employing a gas actuated firing mechanism, the rounds shall be fired as follows: (1) one-third at -65°F, (2) one-third at +70°F, and (3) one-third at +200°F. Conditioning time and method of firing shall be as specified in 4.7.2. To verify the requirement of 3.6.8, gradually apply 400 psig, and hold at 400 psig for one minute. Remove the 400 psig from the mechanism and apply 600 psig at the specified rate of application determined during DVT (see 3.15.2). In addition to the requirements of 4.3, record time from pressure application to firing pin first motion.

4.7.6 Actuation test. To check the requirements of 3.6.9 and 3.6.13 suddenly apply the required CAD actuation pressure at the specified rate of application determined during DVT (see 3.15.2). The rounds shall be fired as follows: (1) one-third at -65°F, (2) one-third at +70°F, and (3) one-third at +200°F. Conditioning time and method shall be as specified in 4.7.2. In addition to the requirements of 4.3, record the time from pressure application to shear pin failure.

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4.7.7 High altitude tests (80,000 ft altitude and -65°F). The CADs shall be conditioned at a temperature of -65°F, until temperature equilibrium is reached. The absolute pressure in the chamber shall then be reduced to 0.82 inch of mercury (corresponding to an altitude of 80,000 feet above sea level). These conditions shall be maintained for one hour. At the end of this time, the CADs are to be fired in the conditioned state. Test data shall be recorded.

4.7.8 Submerged operation. CADs shall be fired while submerged in water to a depth of 100 feet or, as an alternate method, CADs may be tested for submerged operation in a pressurized water chamber at a pressure corresponding to that experienced at a depth of 100 feet and functionally tested.

4.7.9 Iced conditions. The temperature of the CAD shall be reduced to -65°F and stabilized. The CAD shall be transferred to an atmosphere having a temperature of approximately +100°F and a relative humidity of approximately 90 percent until all evidence of frost has disappeared. Retaining all the condensation practicable, the temperature of the CAD shall again be stabilized at -65°F and then fired. If firing is conducted outside the conditioning chamber, it shall be done within three minutes after removal from the chamber.

4.7.10 Structural integrity tests.

4.7.10.1 Locked shut firings. Two CADs shall be conditioned, one at a temperature of -65°F and the other at a temperature of +200°F and fired under locked shut conditions to demonstrate conformance to 3.6.5.1. It is not required that the CAD be operable after test. Conditioning time and method of firing shall be as specified in 4.7.2.

4.7.10.2 No load firings. With the CADs subjected to the no load condition, the CAD firings shall be conducted at -65°F and +200°F. The devices shall meet the requirements of 3.6.5.2. Conditioning time and method shall be as specified in 4.7.2.

4.8 Packaging, packing, and marking inspection. Inspection of packaging and packing of CADs and container markings shall be made to determine compliance with section 5. See 3.17 and 6.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging and packing of CADs and inner and outer container markings shall be subject to the approval of the cognizant design agency. See 3.9.5.1 and 6.4.

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6. NOTES

6.1 Intended use. The CADs covered by this specification are intended to be used in conjunction with cartridges and/or ballistic energy sources.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing and marking requirements. See section 5.

c. Quantity per pack.

d. Data requirements (see 3.9).

6.3 Definitions.

6.3.1 "Sealed-in" type cartridge actuated device. Cartridge actuated devices in which the propellant, explosive, or pyrotechnic components become an integral part of the device and which by design are not to be disassembled and reassembled for inspection are termed "sealed-in" cartridge actuated devices.

6.3.2 "Precocked" type firing mechanism. A mechanical firing mechanism which contains sufficient stored energy in its safetied position for primer initiation is termed a "precocked" firing mechanism.

6.3.3 Cognizant design agency. Unless the contract specifically indicates otherwise, all reference herein to "the cognizant design agency" are defined as referring to the Naval Air Systems Command.

6.3.4 Locked-shut condition. A locked-shut firing is defined as a CAD firing simulating the most adverse conditions for the containment of ballistic gas pressure produced when the device fails to operate normally in the intended manner. The exact nature of this operational anomaly (failure of a piston to move, mechanism to operate, port to vent, etc.) is dependent upon the particular CAD design.

6.4 Packaging, packing, and marking. Specify the appropriate method of preservation in accordance with MIL-P-116, including contact preservative required or without contact preservative as appropriate. Packing shall be specified by level A and level C and meeting the requirements of MIL-STD-794. Marking of packaging and packing shall be in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-178 as applicable.

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6.5 Contractor furnished equipment (CFE).

6.5.1 Quality assurance levels for CFE qualification and production lot acceptance testing shall be subject to the approval of the cognizant design agency.

6.5.2 Production lot size for CADs supplied by a contractor for initial service use shall be subject to the approval of the cognizant design agency.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-D-23615B(AS) DESIGN AND EVALUATION OF CARTRIDGE ACTUATED DEVICES

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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